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1  # Full LC Oscillator Report
2
3  ## 1. Optimization Simulation Setup
4  | Item | Value |
5  |-----|-----|
6  | VDD (V) | 1.8 |
7  | Target Frequency Fo (Hz) | 5 GHz |
8  | Frequency Offset ΔF (Hz) | 1 MHz |
9  | Temperature (°C) | 27 |
10 | Process Corner | NA |
11 | Inductor Width (μm) | 6 |
12 | Inductor Spacing (μm) | 3 |
13 | Population Size | 50 |
14 | Max Generations | 30 |
15 | Stall Generation Limit | 10 |
16 | Crossover Probability (pcross) | 0.9 |
17 | Mutation Probability (pmut) | 0.2 |
18 | Acceptance Tolerance | -1e9 |
19 | Stop Cost Threshold | acceptTol |
20 | PVT Corners | tt, ss, ff, sf, fs |
21 | VDD Sweep (V) | 1.62, 1.8, 1.98 |
22 | Temperature Sweep (°C) | -40, 0, 27, 70, 85 |
23
24 ---
25
26 ## 2. Best Solution Summary
27 | Parameter | Value |
28 |-----|-----|
29 | Generation | 11 |
30 | Index | 33 |
31 | Total Index | 584 |
32 | Fosc | 4.9959 GHz |
33 | Power | 9.148 mW |
34 | Vpk | 1.443 V |
35 | Vrms | 0.722 V |
36 | PN @100k | -92.97 dBc/Hz |
37 | PN @1M | -119.09 dBc/Hz |
38 | FOM @100k | -177.33 |
39 | FOM @1M | -183.45 |
40
41 ---
42
43 ## 3. Constraint Verification
44 | Metric | Target | Result | Status |
45 |-----|-----|-----|-----|
46 | Fosc | 4.9-5.1 GHz | 4.996 GHz | PASS |
47 | Power | ≤10 mW | 9.15 mW | PASS |
48 | Vpk | 0.5-1.5 V | 1.44 V | PASS |
49 | PN @1M | ≤ -100 dBc/Hz | -119.09 | PASS |
50
51 ---
52
53 ## 4. PVT Summary (Extremes)
54 - Frequency: **4.78 - 5.24 GHz**
55 - Power: **4.01 - 18.03 mW**
56 - Vpk: **1.06 - 1.87 V**
57 - PN @1M: **-120.30 to -111.98 dBc/Hz**
58 - FOM @1M: **-184.58 to -177.77**
59
60 ---
61
62 ## 5. Monte Carlo Summary
63 | Metric | Mean | Sigma |
64 |-----|-----|-----|
65 | Fosc | 4.996 GHz | 29.59 MHz |
66 | Power | 9.17 mW | 0.47 mW |
67 | PN @1M | -115.9 | 0.29 |
68 | Vpk | 1.443 V | 0.017 V |
69 | FOM_1M | -180.3 | 0.2849 |

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73 ## 6. Yield Report
74 - Total Samples: 1000
75 - Passing Samples: 958
76 - **Yield: 95.80%**
77
78 ---
79
80 ## 7. Transient (X-Ray) Results
81 - Oscillation Frequency: **5.002 GHz**
82 - Differential Swing: **2.744 Vpp**
83 - RMS Differential: **1.008 V**
84
85 ---
86
87 ## 8. Runtime Summary
88 - Total Runtime: **2.78 hours**
89
```